

13. Patterns (Part II) - Advanced

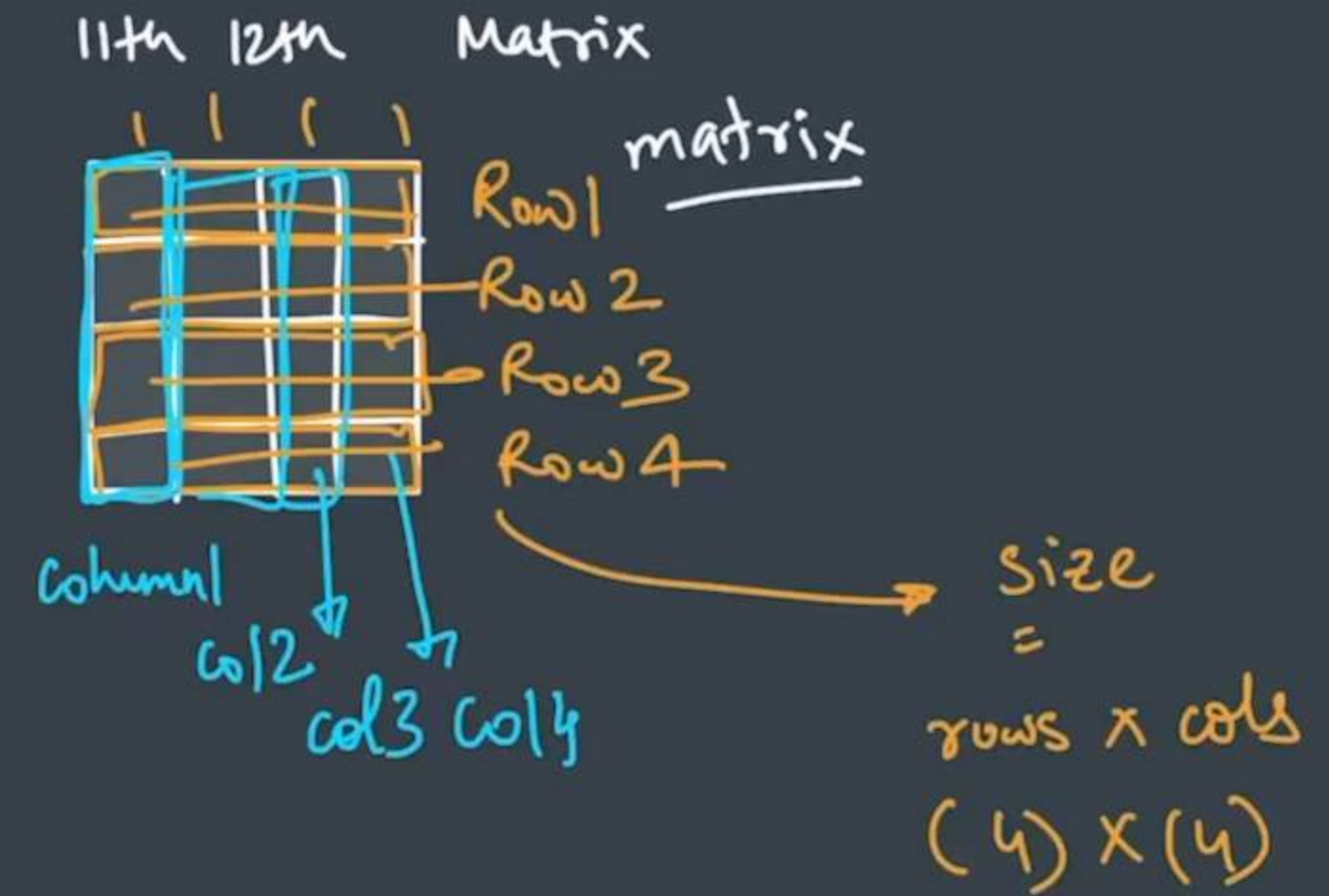
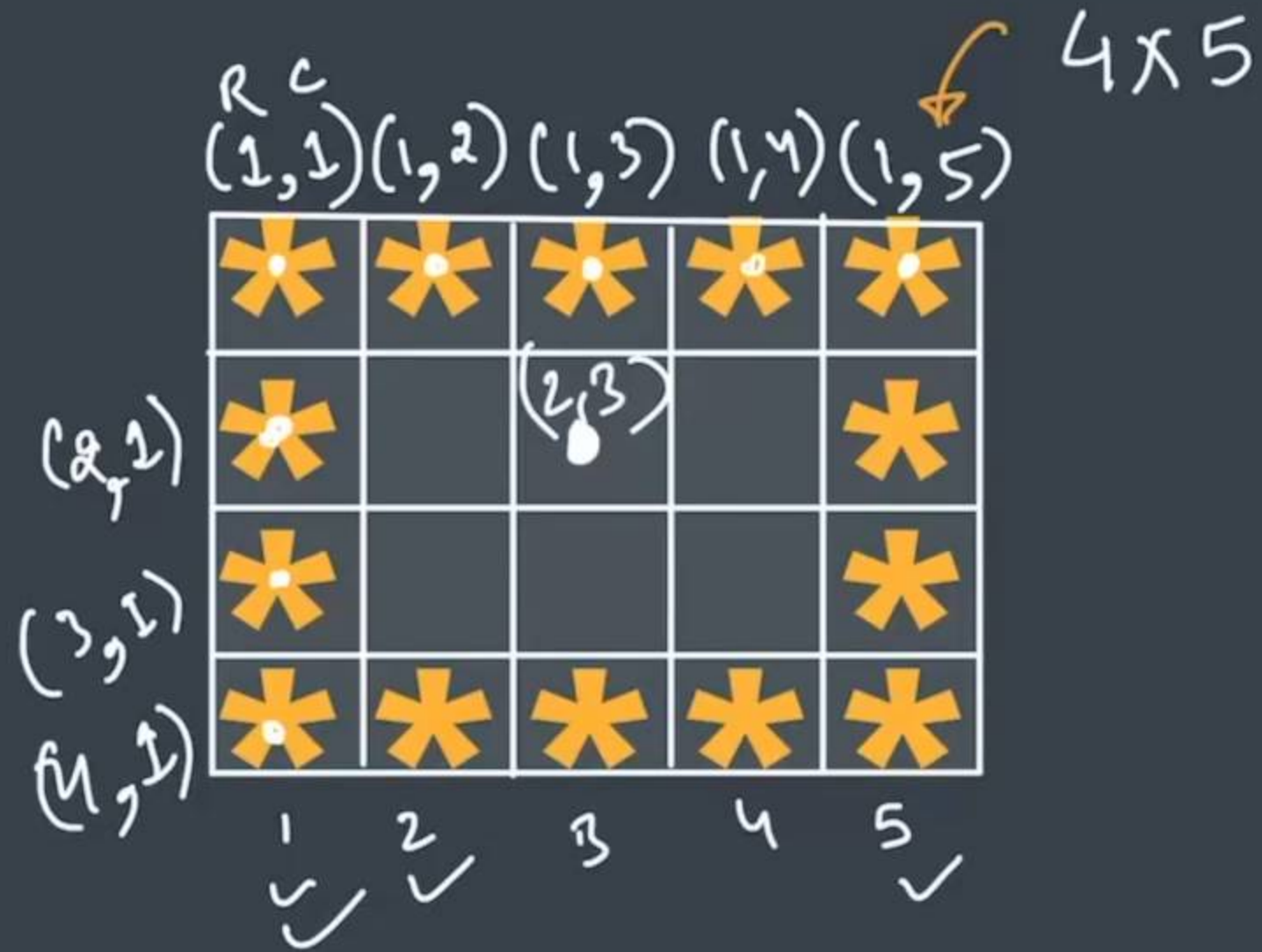


-  **Hollow Rectangle Pattern**
✓ video 11:26
-  **Dry Run - Hollow Rectangle Pattern**
• video 15:33
-  **Inverted & Rotated Half Pyramid**
video 12:42
-  **Inverted Half Pyramid Pattern with Numbers**
video 08:36
-  **Floyd's Triangle Pattern**
video 06:41
-  **0-1 Triangle Pattern**
video 09:25
-  **Butterfly Pattern**
video 07:24
-  **Butterfly Pattern Code (explanation continued as well)**
video 07:46
-  **Solid Rhombus Pattern**
video 07:47
-  **Hollow Rhombus Pattern**
video 07:51
-  **Diamond Pattern**
video 09:56

Print **HOLLOW RECTANGLE** pattern



Print **HOLLOW RECTANGLE** pattern



armankhan082020@gmail.com



Users > shradhakhapra > Patterns.java > Patterns > main(String[])

```
1 public class Patterns {
2     public static void hollow_rectangle(int totRows, int totCols) {
3         //outer loop
4         for(int i=1; i<=totRows; i++) {
5             //inner - columns
6             for(int j=1; j<=totCols; j++) {
7                 //cell - (i,j)
8                 if(i == 1 || i == totRows || j == 1 || j == totCols) {
9                     //boundary cells
10                    System.out.print("*");
11                } else {
12                    System.out.print(" ");
13                }
14            }
15            System.out.println();
16        }
17    }
18    public static void main(String args[]) {
19        |
20    }
21 }
22
```

Run | Debug



totRows = 4

totCols = 5

Print **HOLLOW RECTANGLE** pattern

DRY RUN

i = 1

j = 1 2 3 4 5 6

i = 2

j = 1 2 3 4 5 6

i = 3

j = 1 2 3 4 5 6

i = 4

j = 1 2 3 4 5 6

i = 5

```
//outer loop
for(int i=1; i<=totRows; i++) {
    //inner - columns
    for(int j=1; j<=totCols; j++) {
        //cell - (i,j)
        if(i == 1 || i == totRows || j == 1 || j == totCols) {
            //boundary cells
            System.out.print("*");
        } else {
            System.out.print(" ");
        }
    }
    System.out.println();
}
```

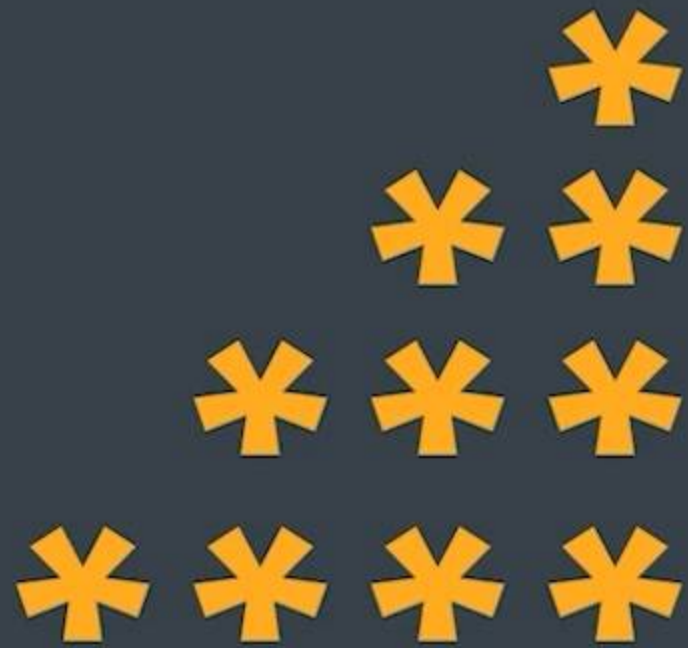
APNA
COLLEGE

output

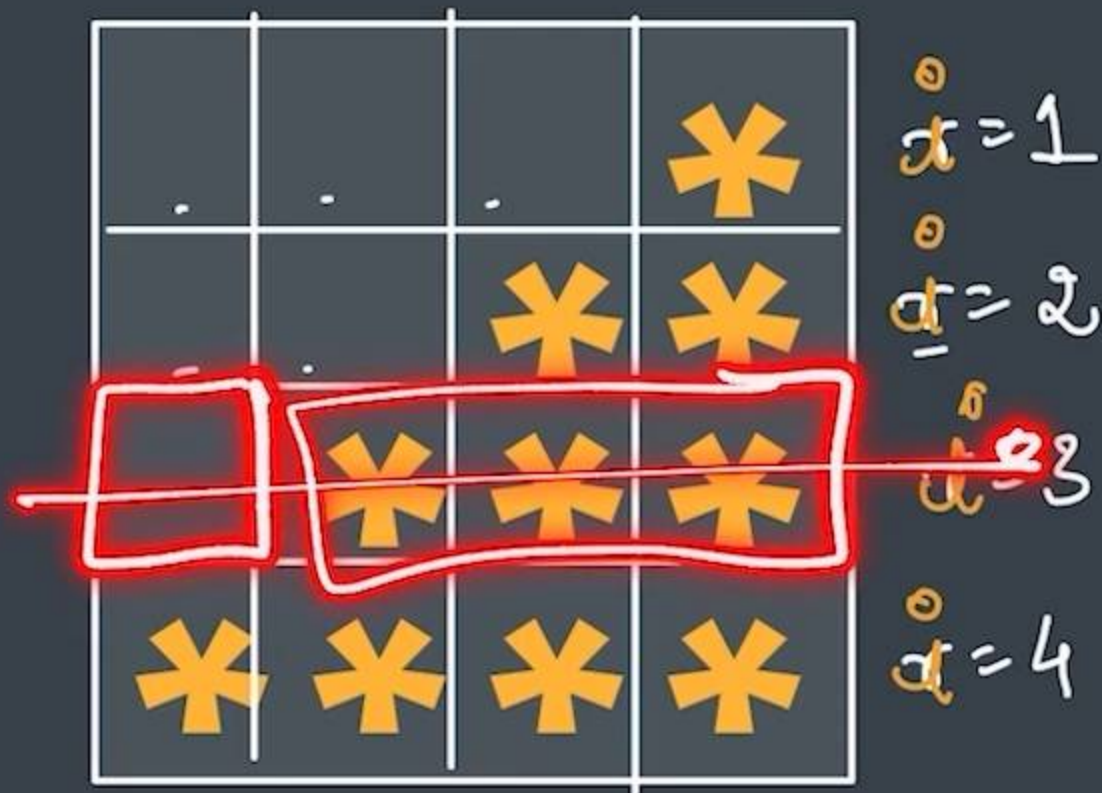
```
* * * * *
*   _   _   _   *
*   _   _   _   *
* * * * *
```



INVERTED & ROTATED HALF-PYRAMID



INVERTED & ROTATED HALF-PYRAMID



for (int i=1; i<=n; i++) {

→ // spaces
 for (int j=1; j<=n-i; j++) {

}

// stars

for (int j=1; j<=i; j++) {
 " "*j
 "

}

→ next line

①
}

②

row no. (i) ⇒ stars
 line

spaces ⇒ $4 - i$
 $= n - i$



Users > shradhakhapra > Patterns.java > Patterns > main(String[])

```
19 public static void inverted_rotated_half_pyramid(int n) {
20     //outer
21     for(int i=1; i<=n; i++) {
22         //spaces
23         for(int j=1; j<=n-i; j++) {
24             System.out.print(" ");
25         }
26
27         //stars
28         for(int j=1; j<=i; j++) {
29             System.out.print("*");
30         }
31
32         System.out.println();
33     }
34 }
```

Run | Debug

```
35 public static void main(String args[]) {
36     hollow_rectangle(10, 7);
37
38 }
39
40 }
```



armankhan082020@gmail.com

Users > shradhakhapra > Patterns.java > Patterns > main(String[])

```
30     }
31
32     System.out.println();
33 }
34
Run | Debug
35 public static void main(String args[]) {
36     //hollow_rectangle(10, 7);
37     inverted_rotated_half_pyramid(4);
38 }
39 }
40
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

shradhakhapra@Shradhas-Air ~ % java Patterns.java

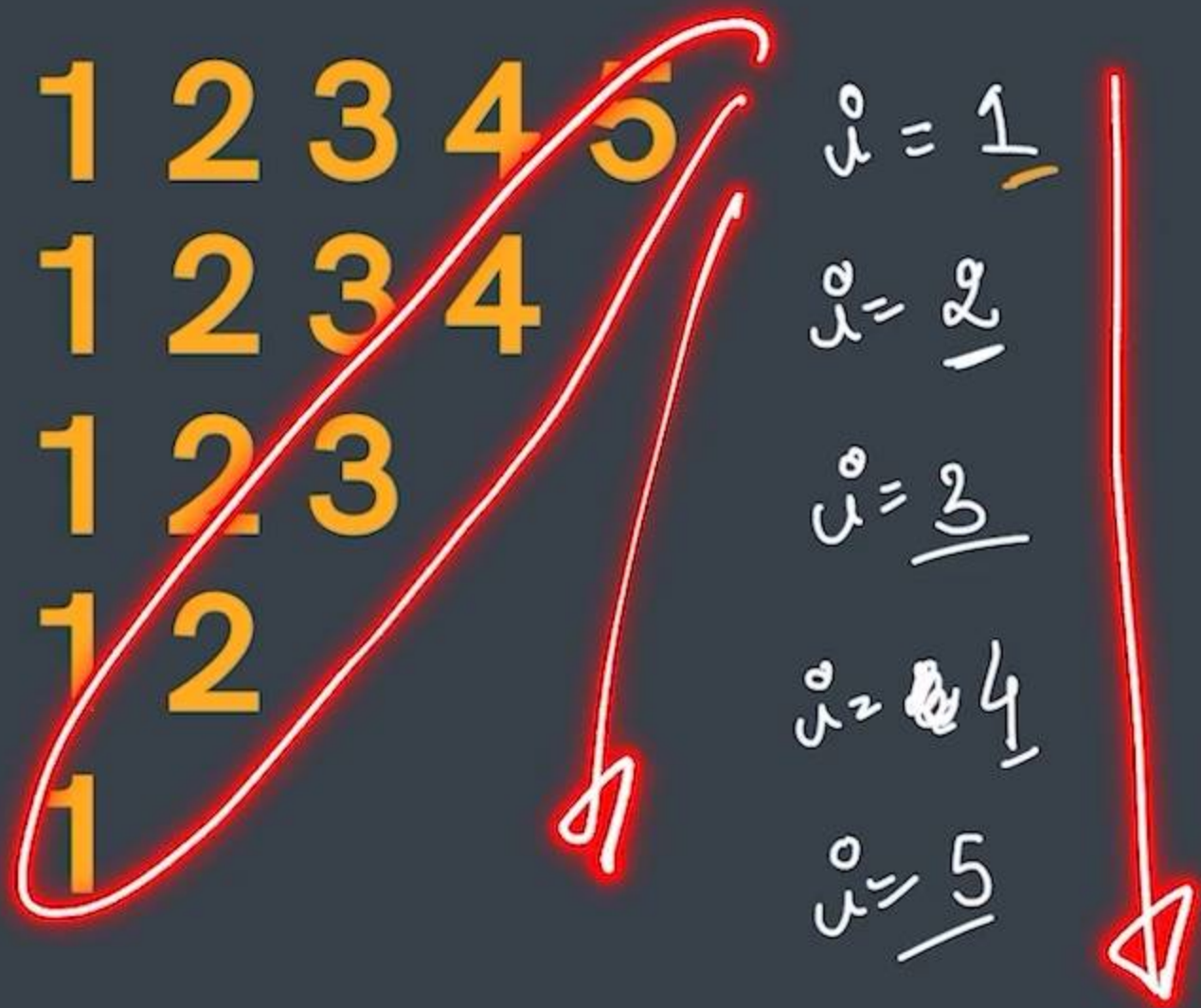
```
*
**
***
****
```

shradhakhapra@Shradhas-Air ~ %



$$n = 5$$

INVERTED HALF-PYRAMID with Numbers



$$i = 1$$

$$i = 2$$

$$i = 3$$

$$i = 4$$

$$i = 5$$

$$1 \text{ to } 5 \quad n - i + 1 = 5 - 1 + 1 = 5$$

$$1 \text{ to } 4 \quad n - i + 1 = 5 - 2 + 1 = 4$$

$$1 \text{ to } 3 \quad n - i + 1 = 5 - 3 + 1 = 3$$

$$1 \text{ to } 2 \quad n - i + 1 = 5 - 4 + 1 = 2$$

$$1 \text{ to } 1 \quad n - i + 1 = 5 - 5 + 1 = 1$$

① lines — outer loop
↳ 5 times (n)
1 to 5

② for (int $i = 1$; $i \leq n$; $i++$) {
 for (int $j = 1$; $j \leq n - i + 1$; $j++$) {
 // print number
 }
}



$$n = 5$$

INVERTED HALF-PYRAMID with Numbers

1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

$$i = 1$$

$$i = 2$$

$$i = 3$$

$$i = 4$$

$$i = 5$$

$$1 \text{ to } 5 \quad n - i + 1 = 5 - 1 + 1 = 5$$

$$1 \text{ to } 4 \quad n - i + 1 = 5 - 2 + 1 = 4$$

$$1 \text{ to } 3 \quad n - i + 1 = 5 - 3 + 1 = 3$$

$$1 \text{ to } 2 \quad n - i + 1 = 5 - 4 + 1 = 2$$

$$1 \text{ to } 1 \quad n - i + 1 = 5 - 5 + 1 = 1$$

① lines — outer loop
↳ 5 times (n)
1 to 5

② for (int i = 1; i <= n; i++) {
 for (int j = 1; j <= n - i + 1; j++)
 }
}



Users > shradhakhapra > Patterns.java > Patterns > inverted_half_pyramid_withNumbers(int)

APNA
COLLEGE

```
32      System.out.println();
33    }
34  }
35
36  public static void inverted_half_pyramid_withNumbers(int n) {
37      for(int i=1; i<=n; i++) {
38          //inner - numbers
39          for(int j=1; j<=n-i+1; j++) {
40              System.out.print(j);
41          }
42          System.out.println();
43      }
44  }
45
46  Run | Debug
47  public static void main(String args[]) {
48      //hollow_rectangle(10, 7);
49      //inverted_rotated_half_pyramid(7);
50      inverted_half_pyramid_withNumbers(5);
51  }
```

armankhan082020@gmail.com



FLOYD'S Triangle

```
int n=5; int counter=1;
```

```
for (int i=1; i<=n; i++) {
```

```
    for (int j=1; j<=i; j++)
```

```
    {
```

```
        counter → print  
        counter++
```

```
    }
```

lines
APNA
COLLEGE
counter
print
inner

1 → i=1 (1)
2 3 → i=2 (2)
4 5 6 → i=3 (3)
7 8 9 10 → i=4 (4)
11 12 13 14 15 → i=5 (5)



Users > shradhakhapra > Patterns.java > Patterns > floyds_triangle(int)

```
39         for(int j=1; j<=n-1+1; j++) {
40             System.out.print(j);
41         }
42         System.out.println();
43     }
44 }
45
46 public static void floyds_triangle(int n) {
47     //outer
48     int counter = 1;
49     for(int i=1; i<=n; i++) {
50         //inner - how many times will counter be printed
51         for(int j=1; j<=i; j++) {
52             System.out.print(counter+" ");
53             counter++;
54         }
55         System.out.println();
56     }
57 }
58
59 Run | Debug
60 public static void main(String args[]) {
61     //hollow_rectangle(10, 7);
62     //inverted_rotated_half_pyramid(7);
63     inverted_half_pyramid_withNumbers(5);
64 }
```



Users > shradhakhapra > Patterns.java > Patterns > main(String[])

```
57 }  
Run | Debug  
58 public static void main(String args[]) {  
59     //hollow_rectangle(10, 7);  
60     //inverted_rotated_half_pyramid(7);  
61     //inverted_half_pyramid_withNumbers(5);  
62     floyds_triangle(4);  
63 }  
64 }
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

shradhakhapra@Shradhas-Air ~ % java Patterns.java

```
1  
2 3  
4 5 6  
7 8 9 10  
11 12 13 14 15
```

shradhakhapra@Shradhas-Air ~ % java Patterns.java

```
1  
2 3  
4 5 6  
7 8 9 10
```

shradhakhapra@Shradhas-Air ~ %



0-1 Triangle

$n=5$

(row + col)

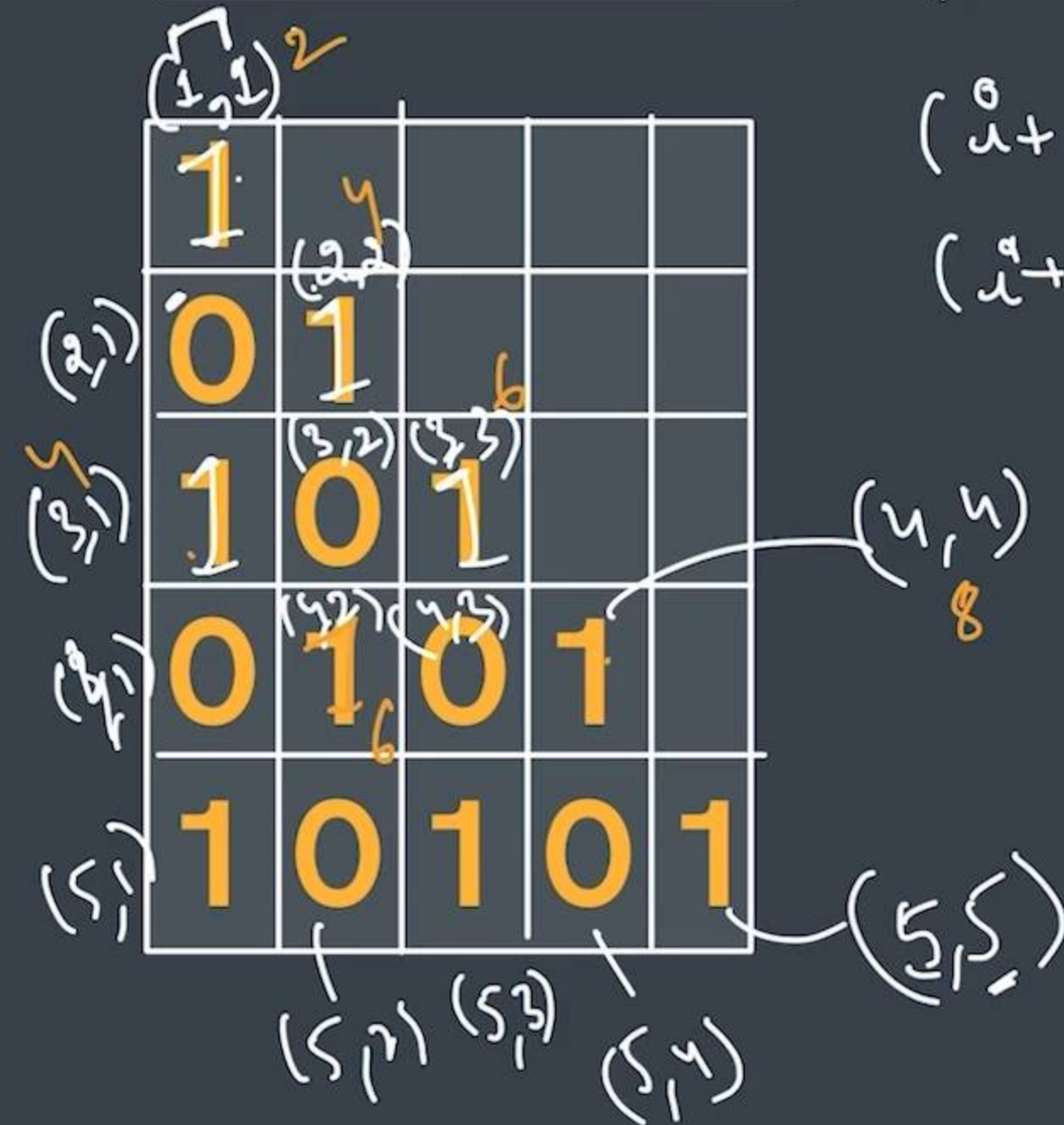
$(i+j) \rightarrow \text{even} \rightarrow "1"$

$(i+j) \rightarrow \text{odd} \rightarrow "0"$

① outer loop
↳ n lines

for (int i=1; i<=n; i++)

② each line



i - outer loop Row
 j - inner loop Col

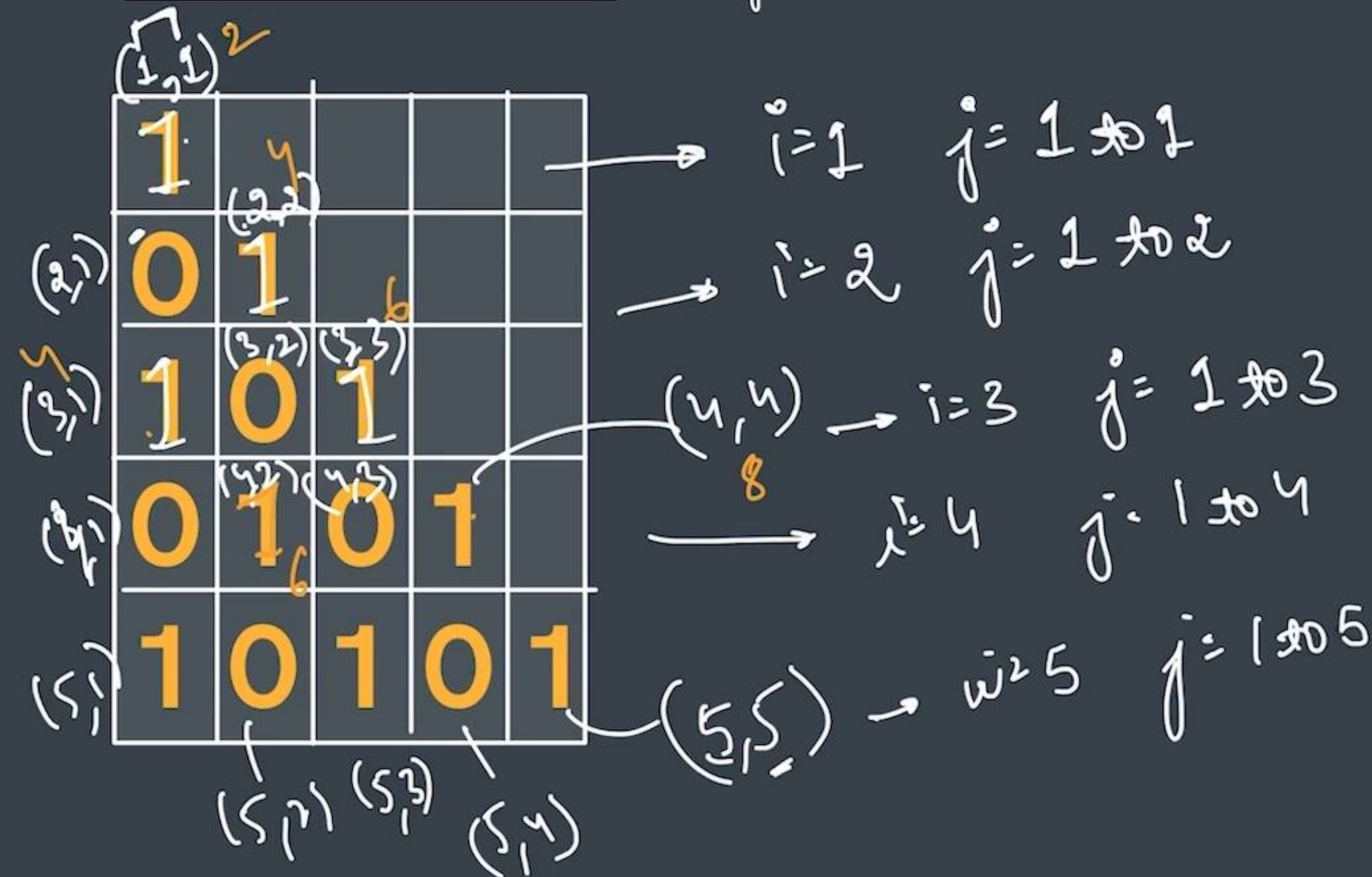


0-1 Triangle

(row + col)

$(i+j) \rightarrow \text{even} \rightarrow 1$

$(i+j) \rightarrow \text{odd} \rightarrow 0$



① outer loop
↳ n lines

for (int i=1; i<=n; i++)

② each line

for (int j=1; j<=i; j++)



Users > shradhakhapra > Patterns.java > Patterns > zero_one_triangle(int)

```
55     System.out.println();
56 }
57 }
58
59 public static void zero_one_triangle(int n) {
60     for(int i=1; i<=n; i++) {
61         for(int j=1; j<=i; j++) {
62             if( (i+j) % 2 == 0) { // even
63                 System.out.print("1");
64             } else {
65                 System.out.print("0");
66             }
67         }
68
69         System.out.println();
70     }
71 }
72
73 public static void main(String args[]) {
74     //hollow_rectangle(10, 7);
75 }
```

Run | Debug

armankhan082020@gmail.com

PROBLEMS

1

OUTPUT

TERMINAL

DEBUG CONSOLE

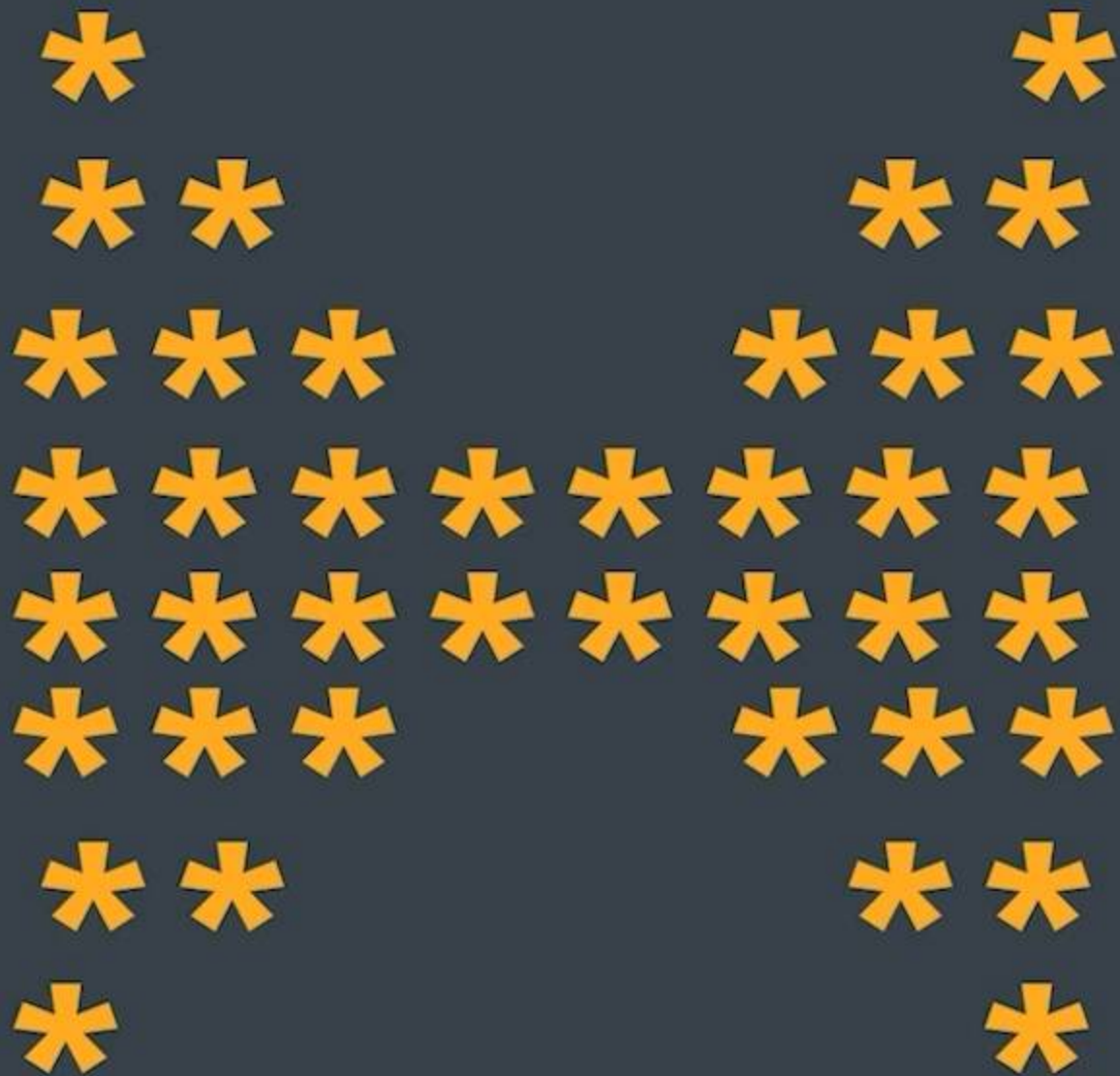
10101

shradhakhapra@Shradhas-Air ~ %

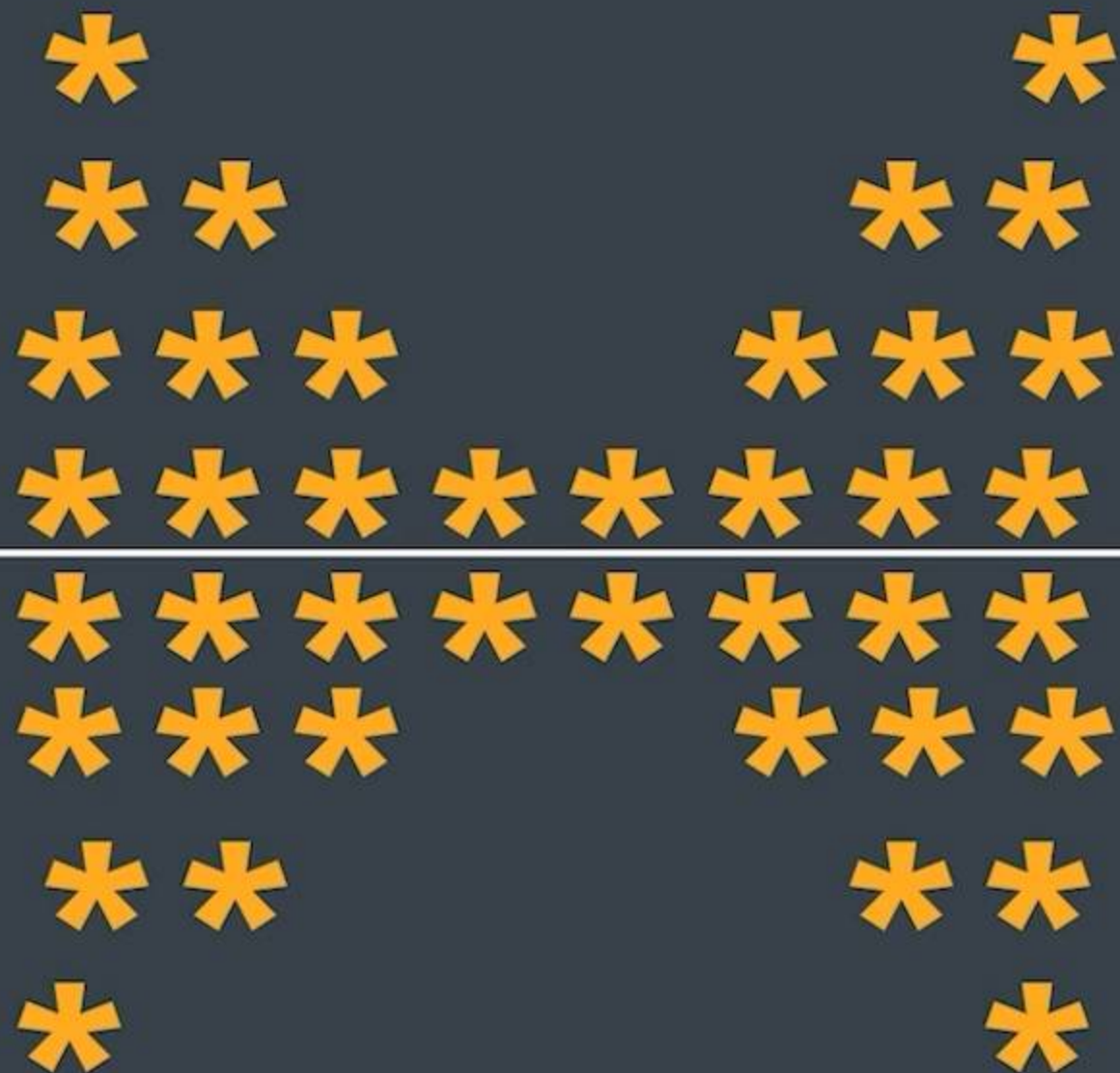


η=

BUTTERFLY Pattern



BUTTERFLY Pattern



$n=4$

1st
Half

①

outer

for (int i=1; i<=n; i++)

$(n-i)$

②

line

↓

stars

(i)

+ spaces + stars
 $2 \times (n-i)$ (i)

$i=1$
 $i=2$
 $i=3$
 $i=4$

$sp = 6 = 2 \times 3$
 $sp = 4 = 2 \times 2$
 $sp = 2 = 2 \times 1$
 $sp = 0 = 2 \times 0$

$(n-i)$
 $4-1=3$
 $4-2=2$
 $4-3=1$
 $4-4=0$

spaces = $2 \times (n-i)$



```
72
73     public static void butterfly(int n) {
74         //1st half
75         for(int i=1; i<=n; i++) {
76             //stars - i
77             for(int j=1; j<=i; j++) {
78                 System.out.print("*");
79             }
80
81             //spaces - 2*(n-i)
82             for(int j=1; j<=2*(n-i); j++) {
83                 System.out.print(" ");
84             }
85
86             //stars - i
87             for(int j=1; j<=i; j++) {
88                 System.out.print("*");
89             }
90         }
91     }
92
93     public static void main(String args[]) {
94         //hollow_rectangle(10, 7);
95         //inverted_rotated_half_square(7);
```



Users > shradhakhapra > Patterns.java > Patterns > butterfly(int)

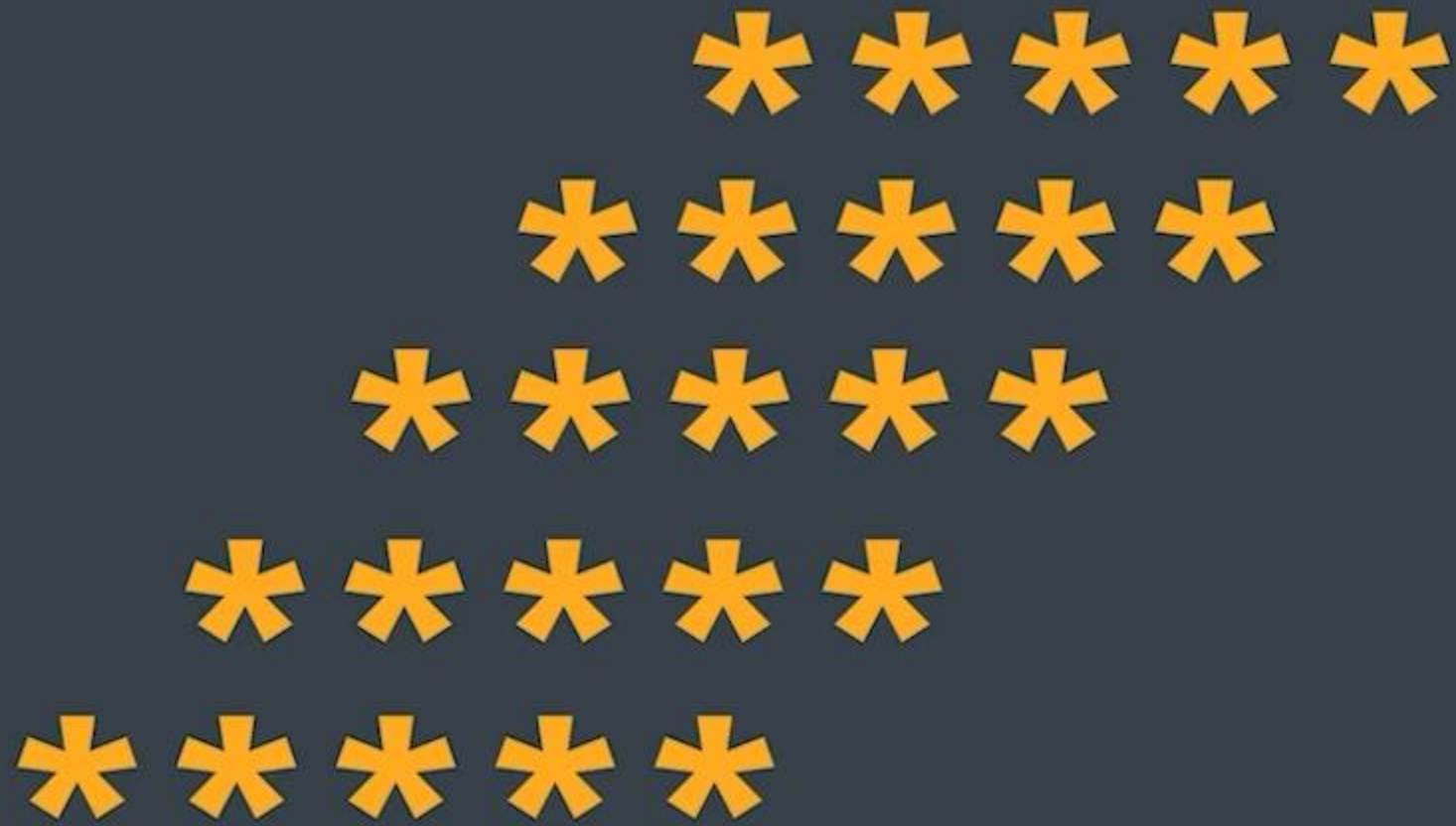
APNA
COLLEGE

```
86 //stars - i
87 for(int j=1; j<=i; j++) {
88     System.out.print("*");
89 }
90 }

91
92 //2nd Half
93 for(int i=n; i>=1; i--) {
94     //stars - i
95     for(int j=1; j<=i; j++) {
96         System.out.print("*");
97     }
98
99     //spaces - 2*(n-i)
100    for(int j=1; j<=2*(n-i); j++) {
101        System.out.print(" ");
102    }
103
104    //stars - i
105    for(int j=1; j<=i; j++) {
106        System.out.print("*");
107    }
108 }
109 }
```



SOLID RHOMBUS



SOLID RHOMBUS

$n = 5$

$stars = n$ ($j = 1$ to n)
 $spaces = (n - i)$ ($j = 1$ to n)

APNA
COLLEGE

$$n - i = 5 - 1 = 4$$

$$spaces = 4 \quad stars = 5$$

$$sp = 3 \quad (5 - 2) = 3 \quad st = 5$$

$$sp = 2 \quad (5 - 3) = 2 \quad st = 5$$

$$sp = 1 \quad (5 - 4) = 1 \quad st = 5$$

$$sp = 0 \quad (5 - 5) = 0$$



$i = 1$
 $i = 2$
 $i = 3$
 $i = 4$
 $i = 5$

① outer loop
↳ $n = 5$
`for (int i = 1; i <= 5; i++)`

② `line(i)`
spaces + stars



```
112
113 public static void solid_rhombus(int n) {
114     for(int i=1; i<=n; i++) {
115         //spaces
116         for(int j=1; j<=(n-i); j++) {
117             System.out.print(" ");
118         }
119
120         //stars
121         for(int j=1; j<=n; j++) {
122             System.out.print("*");
123         }
124
125         System.out.println();
126     }
127 }
128
```

arman Khan082020@gmail.com

Run | Debug

```
129 public static void main(String args[]) {
130     //hollow_rectangle(10, 7);
131     //inverted_rotated_half_pyramid(7);
132     //inverted_half_pyramid_withNumbers(5);
133     //floyds_triangle(4);
134     //zero_one_triangle(5);
135     butterfly(5);

```



Users > shradhakhapra > Patterns.java > Patterns > main(String[])

Run | Debug

```
129 public static void main(String args[]) {
130     //hollow_rectangle(10, 7);
131     //inverted_rotated_half_pyramid(7);
132     //inverted_half_pyramid_withNumbers(5);
133     //floyds_triangle(4);
134     //zero_one_triangle(5);
135     //butterfly(5);
136     solid_rhombus(5);
137 }
138 }
139
```

armankhan082020@gmail.com

PROBLEMS

1

OUTPUT

TERMINAL

DEBUG CONSOLE

shradhakhapra@Shradhas-Air ~ % java Patterns.java

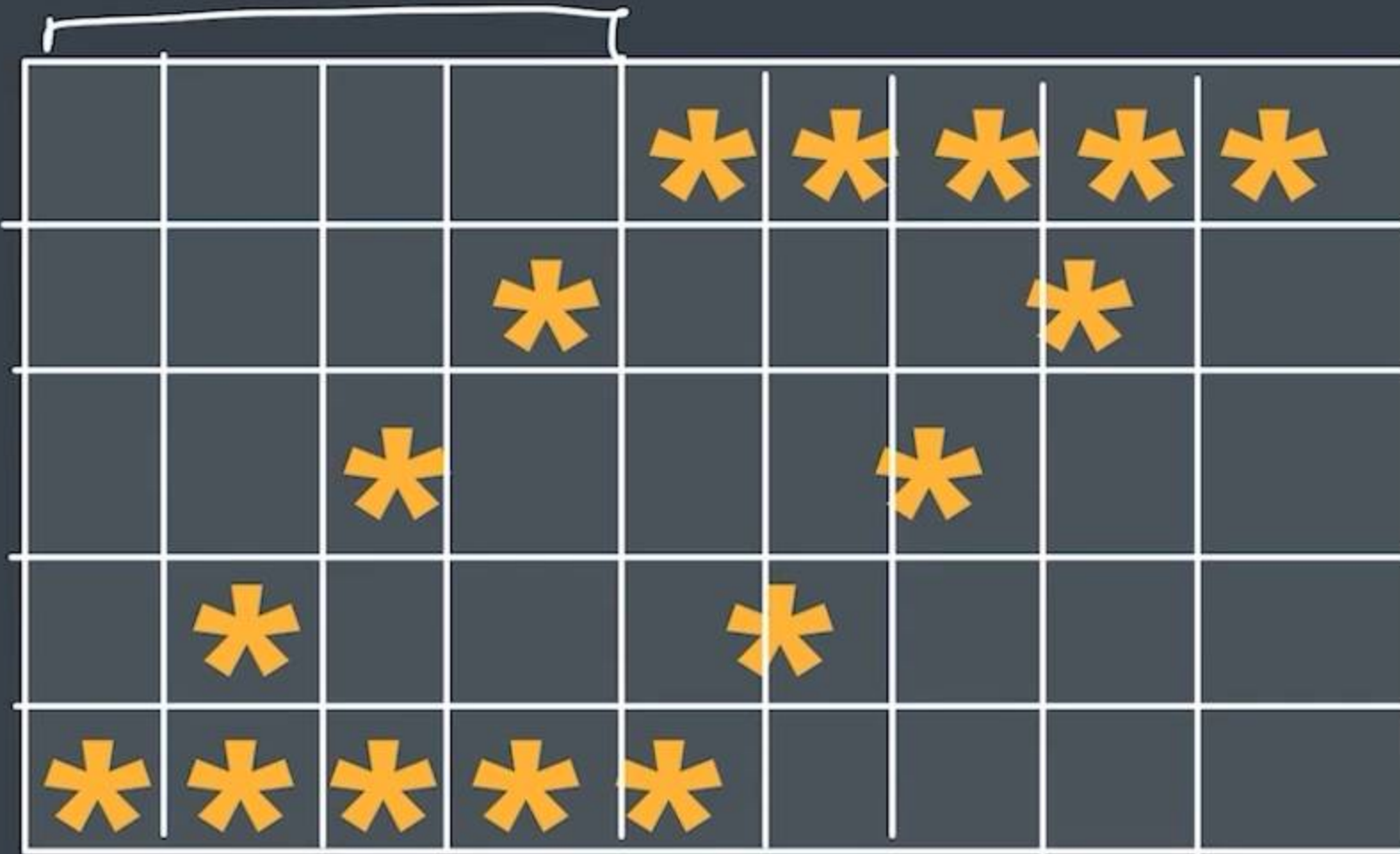
```
*****
*****
*****
*****
*****
```

shradhakhapra@Shradhas-Air ~ %



n=5

HOLLOW RHOMBUS



① outer $\rightarrow$ `for (int i=1 to 5)`

APNA
COLLEGE

② line
spaces + boundary rectangle
 $(n-i)$ $\downarrow$
hollow rectangle



Users > shradhakhapra > Patterns.java > Patterns > hollow_rhombus(int)

```
125     System.out.println();
126 }
127 }
128
129 public static void hollow_rhombus(int n) {
130     for(int i=1; i<=n; i++) {
131         //spaces
132         for(int j=1; j<=(n-i); j++) {
133             System.out.print(" ");
134         }
135
136         //hollow rectangle - stars
137         for(int j=1; j<=n; j++) {
138             if(i == 1 || i == n || j == 1 || j == n) {
139                 System.out.print("*");
140             } else {
141                 System.out.print(" ");
142             }
143         }
144     }
145 }
146
147 public static void main(String args[]) {
```



Users > shradhakhapra > Patterns.java > Patterns > hollow_rhombus(int)

```
139         System.out.print(" * ");
140     } else {
141         System.out.print(" ");
142     }
143 }
144
145 System.out.println();
146 }
147 }
148
149 public static void main(String[] args) {
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

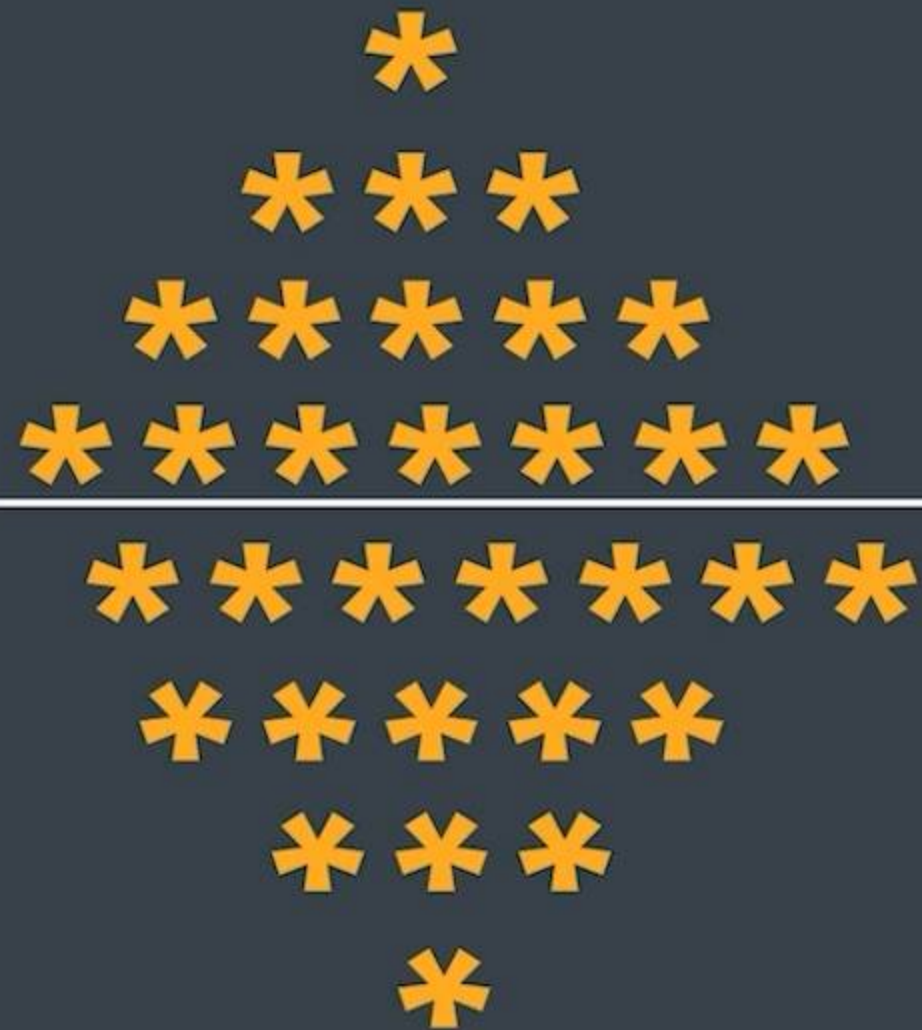
shradhakhapra@Shradhas-Air ~ % java Patterns.java

```
*****
 *   *
 *   *
 *   *
*****
```

shradhakhapra@Shradhas-Air ~ %

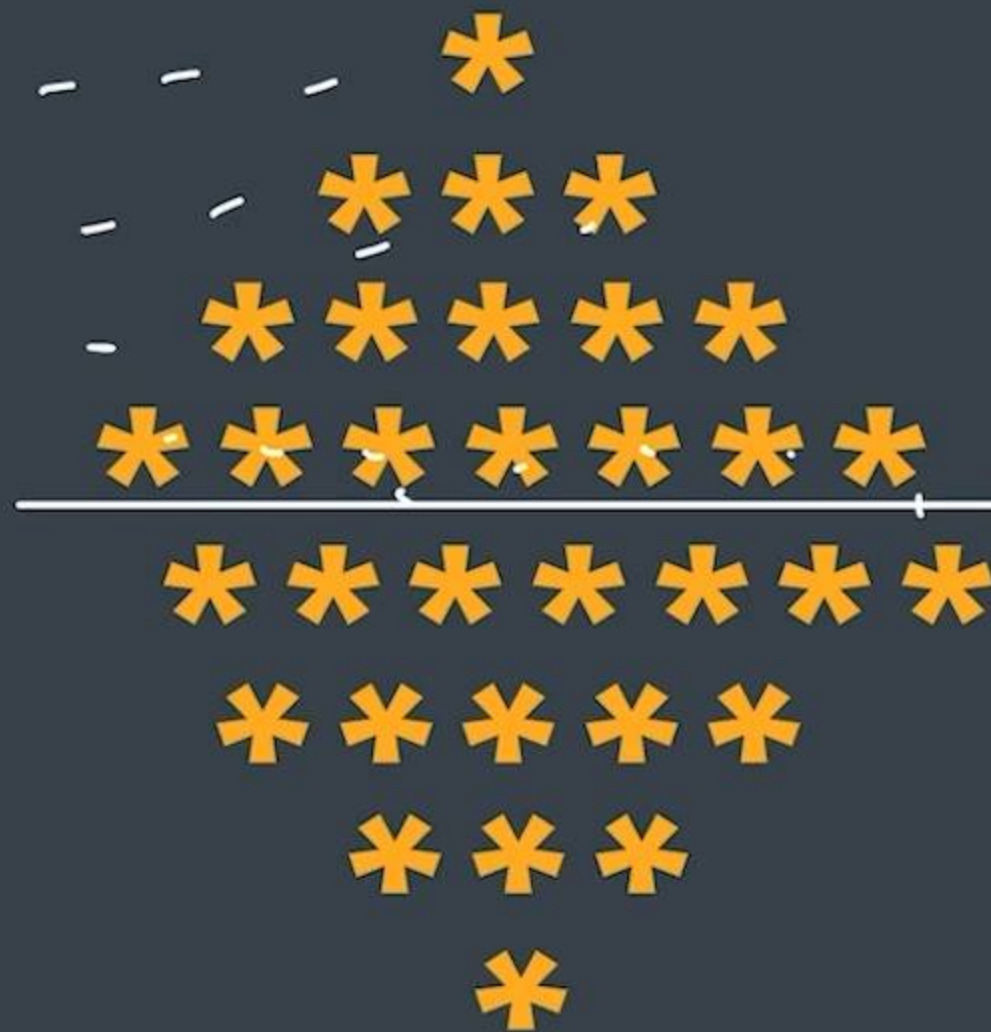


DIAMOND Pattern



DIAMOND Pattern

n=4



i=1
i=2
i=3
i=4

sp = n - i

sp = 3 ⁽⁴⁻¹⁾	st = 1 = 2 ⁱ - 1 = 2(1) - 1 = 1 ✓
sp = 2 ⁽⁴⁻²⁾	st = 3 = 2(2) - 1 = 3 ✓
sp = 1 ⁽⁴⁻³⁾	st = 5 = 2(3) - 1 = 5 ✓
sp = 0 ⁽⁴⁻⁴⁾	st = 7 = 2(4) - 1 = 7 ✓

odd
odd

2i+1	2i-1
2i+1	2i-1
X	✓

- ① outer → 1 to n
for (int i=1; i<=n; i++)
- ② what to print



Users > shradhakhapra > Patterns.java > Patterns > diamond(int)

```
160     }
161
162     System.out.println();
163 }
164
165 //2nd half
166 for(int i=n; i>=1; i--) {
167     //spaces
168     for(int j=1; j<=(n-i); j++) {
169         System.out.print(" ");
170     }
171
172     //stars
173     for(int j=1; j<=(2*i)-1; j++) {
174         System.out.print("*");
175     }
176
177     System.out.println();
178 }
179 }
180
181 public static void main(String args[]) {
182     //hollow_rectangle(10, 7);
```



```
140  
147 }  
148  
149 public static void diamond(int n) {  
150     //1st half  
151     for(int i=1; i<=n; i++) {  
152         //spaces  
153         for(int j=1; j<=(n-i); j++) {  
154             System.out.print(" ");  
155         }  
156  
157         //stars  
158         for(int j=1; j<=(2*i)-1; j++) {  
159             System.out.print("*");  
160         }  
161  
162         System.out.println();  
163     }  
164  
165     //2nd half  
166     for(int i=n; i>=1; i--) {  
167         //spaces  
168         for(int j=1; j<=(n-i); j++) {  
169             System.out.print(" ");  
170     }
```



Users > shradhakhapra > Patterns.java > Patterns > main(String[])

```
174      System.out.print(" * ");
```

```
175      }
```

```
176
```

```
177      System.out.println();
```

```
178      }
```

```
179  }
```

```
180
```

Run | Debug

```
181 public static void main(String args[]) {
```

```
182     //hollow_rectangle(10, 7);
```

```
183     //inverted_rotated_half_pyramid(7);
```

```
184     //inverted_half_pyramid_withNumbers(5);
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

```
shradhakhapra@Shradhas-Air ~ % java Patterns.java
```

```
  I*
```

```
 ***
```

```
*****
```

```
*****
```

```
*****
```

```
*****
```

```
 ***
```

```
  *
```

```
shradhakhapra@Shradhas-Air ~ %
```

